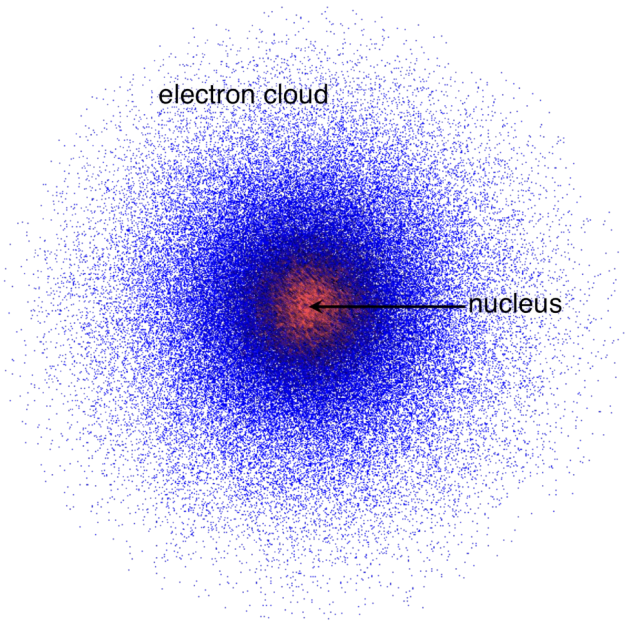
Chapter 3: Vector Calculus

The following subjects will be discussed

- The "del" Operator.
- Gradient, Divergence, Curl and Laplacian operators.

Examples

(3) A cloud of electron with density of $\rho_v = -\frac{3\times10^{-8}}{r^4}\cos^2\varphi$ (C/m^3), exists between r=2 and 5 cm. Find the total charge.



$$\text{Volume Integral} \equiv \iiint_V (density)\, dv$$

Total charge: $Q = \iiint_V -\frac{3\times10^{-8}}{r^4}(\cos^2\varphi)\; r^2 \sin\theta\, dr\, d\theta\, d\varphi$

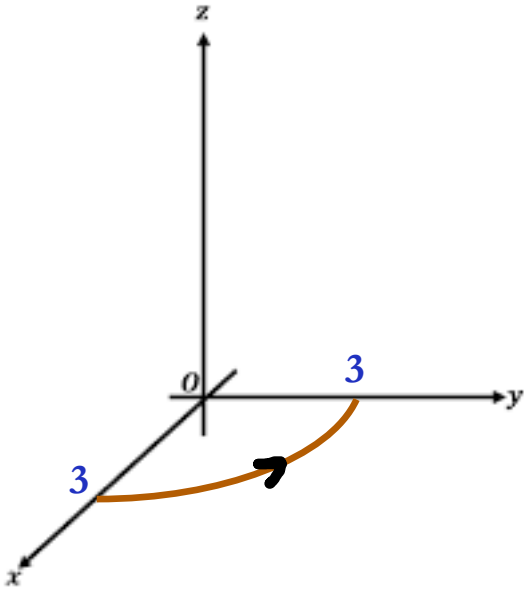
$$Q = -3\times10^{-8}\int_{0.02}^{0.05}\frac{r^2}{r^4}dr\int_0^{\pi}\sin\theta\; d\theta\int_0^{2\pi}(\cos^2\varphi)\; d\varphi$$

$$Q = -3\times10^{-8}\left[\left.\frac{1}{r}\right|_{0.02}^{0.05}\;(-\cos\theta)\Big|_0^{\pi}\;\left(\frac{\varphi}{2}+\frac{\sin 2\varphi}{4}\right)\Big|_0^{2\pi}\right]$$

$$Q = -1.8\pi\times10^{-6} \quad \text{(C)}$$

Examples

(4) Given $\vec{F} = xy\,\vec{a}_x - 2x\,\vec{a}_y$, find the line integral along the indicated path.



Transform to cylindrical coordinate system.

Perform line integral in cylindrical coordinate system (only φ component)

Answer: $-9\left(1+\frac{\pi}{2}\right)$

Vector Differentiation: Del Operator

The "del" operator ($\vec{\nabla}$) is the vector differentiation basic operation. It represents the rate of change with respect to space. It is defined in rectangular, cylindrical and spherical coordinates as follows.

Rectangular coordinate $$\vec{\nabla}= \frac{\partial}{\partial x}\vec{a}_x + \frac{\partial}{\partial y}\vec{a}_y + \frac{\partial}{\partial z}\vec{a}_z$$

Cylindrical coordinate $$\vec{\nabla}= \frac{\partial}{\partial \rho}\vec{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \varphi}\vec{a}_\varphi + \frac{\partial}{\partial z}\vec{a}_z$$

Spherical coordinate $$\vec{\nabla}= \frac{\partial}{\partial r}\vec{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi}\vec{a}_\varphi$$

Being a "vector" operator, it can be used with several possibilities.

	Gradient	Divergence	Curl	Laplacian	Laplacian
	$\vec{\nabla}A$	$\vec{\nabla}\cdot\vec{A}$	$\vec{\nabla}\times\vec{A}$	$\nabla^2 A$	$\vec{\nabla}^2\vec{A}$
Result is:	vector	scalar	vector	scalar	vector

Vector Differentiation: Gradient

The gradient represents the three-dimensional "rate of change" of a given scalar function. So, it generates the rate of change vector.

Rectangular coordinate

$A(x, y, z)$ is a scalar function

$$\vec{\nabla} A = \frac{\partial A}{\partial x}\,\vec{a}_x + \frac{\partial A}{\partial y}\,\vec{a}_y + \frac{\partial A}{\partial z}\,\vec{a}_z$$

Cylindrical coordinate

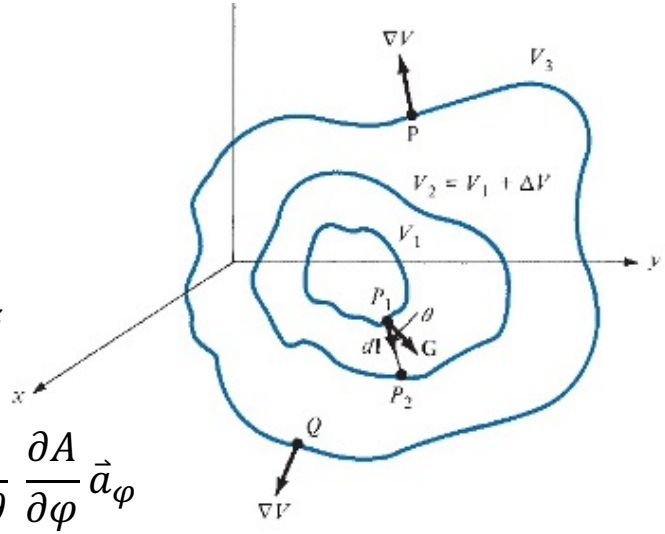
$A(\rho, \varphi, z)$ is a scalar function

$$\vec{\nabla} A = \frac{\partial A}{\partial \rho}\,\vec{a}_\rho + \frac{1}{\rho}\frac{\partial A}{\partial \varphi}\,\vec{a}_\varphi + \frac{\partial A}{\partial z}\,\vec{a}_z$$

Spherical coordinate

$A(r, \theta, \varphi)$ is a scalar function

$$\vec{\nabla} A = \frac{\partial A}{\partial r}\,\vec{a}_r + \frac{1}{r}\frac{\partial A}{\partial \theta}\,\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial A}{\partial \varphi}\,\vec{a}_\varphi$$



Remarks:

- The gradient denotes the direction of greatest change of a scalar function.
- Its magnitude indicates how fast is the function changes in that direction.
- The gradient vector is always normal to equi-valued surfaces.
- The units of the gradient results is: (units of A)/meter.

Example

The electrostatic field intensity is given by the gradient of the electric potential function. Find the electric field intensity at P(1, 1, 0). $\vec{E} = -\vec{\nabla}V$

Electric potential $V(x,y,z) = V_o e^{-x} \sin\frac{\pi y}{4}$ (V)

$$\vec{E} = -\vec{\nabla}V = -\left(\frac{\partial V}{\partial x}\,\vec{a}_x + \frac{\partial V}{\partial y}\,\vec{a}_y + \frac{\partial V}{\partial z}\vec{a}_z\right) = -\left(-V_o e^{-x}\sin\frac{\pi y}{4}\vec{a}_x + \frac{\pi}{4}V_o e^{-x}\cos\frac{\pi y}{4}\vec{a}_y\right)$$

$$\vec{E}_P = \frac{V_o}{e\sqrt{2}}\left(\vec{a}_x - \frac{\pi}{4}\vec{a}_y\right) \quad \text{(V/m)}$$

Electric potential $V(r,\theta,\varphi) = V_o r\cos\theta$ (V)

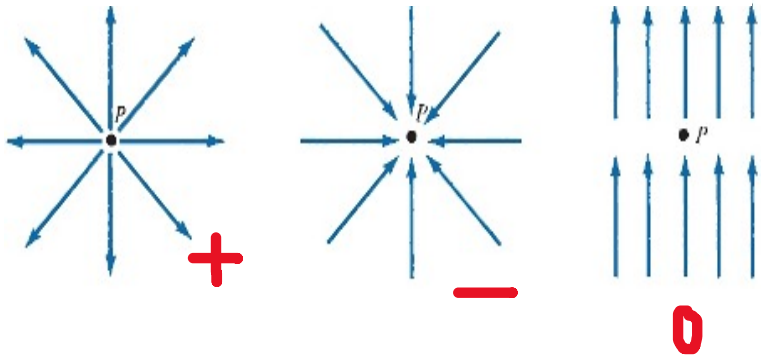
$$\vec{E} = -\vec{\nabla}V = -\left(\frac{\partial}{\partial r}\,\vec{a}_r + \frac{1}{r}\frac{\partial}{\partial\theta}\vec{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\vec{a}_\varphi\right) = -(V_o\cos\theta\,\vec{a}_r - V_o\sin\theta\,\vec{a}_\theta)$$

$$\vec{E}_P = \frac{V_o}{\sqrt{2}}\,\vec{a}_\theta \quad \text{(V/m)} \qquad \text{P(1, 1, 0)} \equiv \text{P}\left(\sqrt{2}, \frac{\pi}{2}, \frac{\pi}{4}\right)$$

Vector Differentiation: Divergence

The divergence operation measures how much a vector field is converging at (accumulating, sink) or diverging from (dispersing, source) a given point.

The result is a scalar quantity that is positive for source and negative for sink.



Rectangular coordinate

$$\vec{\nabla}\cdot\vec{A}=\frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z}$$

$A(x,y,z)$ is a scalar function

Cylindrical coordinate

$$\vec{\nabla}\cdot\vec{A}=\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho A_\rho\right)+\frac{1}{\rho}\frac{\partial A_\varphi}{\partial\varphi}+\frac{\partial A_z}{\partial z}$$

$A(\rho,\varphi,z)$ is a scalar function

Spherical coordinate

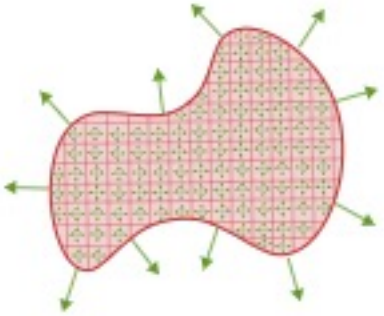
$$\vec{\nabla}\cdot\vec{A}=\frac{1}{r^2}\frac{\partial}{\partial r}(r^2A_r)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}$$

$A(r,\theta,\varphi)$ is a scalar function

Divergence Theorem

$$\int_V \vec{\nabla}\cdot\vec{A}\;dv=\oint_S \vec{A}\cdot\overrightarrow{dS}$$

Sum of all divergence of points inside a volume is equal to the flux out of the volume



Examples

(1) Does the vector field $\vec{D} = 2xy\,\vec{a}_x - 7\,\vec{a}_y + (x^2 + z^2)\,\vec{a}_z$ behave as a source or a sink at P(0, 1, 3)?

$$\vec{\nabla}\cdot\vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \frac{\partial 2xy}{\partial x} - \frac{\partial 7}{\partial y} + \frac{\partial (x^2+z^2)}{\partial z} = 2y + 2z$$

$$\left(\vec{\nabla}\cdot\vec{D}\right)\Big|_P = 8 \qquad \text{(Source)}$$

(2) The magnetic flux density due to a straight thin infinite wire carrying current I, at any distance ρ from the wire, is given by

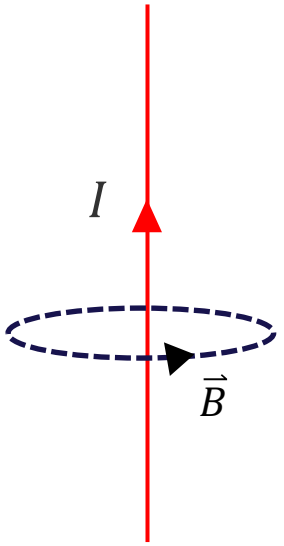
$$\vec{B} = \frac{\mu_o I}{2\pi\rho}\vec{a}_\varphi$$

Describe the field lines.

$$\vec{\nabla}\cdot\vec{B} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho B_\rho) + \frac{1}{\rho}\frac{\partial B_\varphi}{\partial\varphi} + \frac{\partial B_z}{\partial z} = \frac{1}{\rho}\frac{\partial}{\partial\varphi}\left(\frac{\mu_o I}{2\pi\rho}\right) = 0$$

$$\vec{\nabla}\cdot\vec{B} = 0 \quad \Rightarrow \quad \int_V \vec{\nabla}\cdot\vec{A}\,dv = \oint_S \vec{A}\cdot\overrightarrow{dS} = 0$$

Divergence theorem



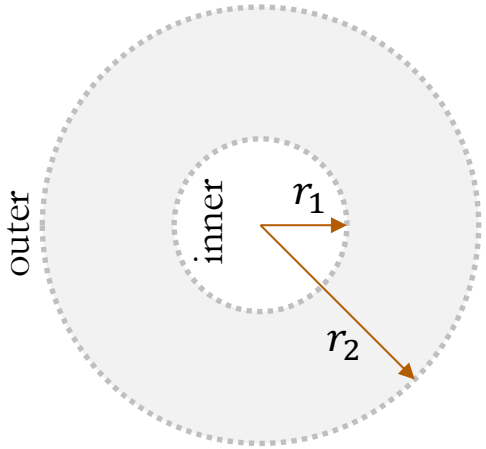
Magnetic field lines go around in circles (solenoidal)

Examples

(3) Verify the divergence theorem of the vector $\vec{F} = kr\,\vec{a}_r$ over the given surface.

$$\vec{\nabla}\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(F_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial F_\varphi}{\partial\varphi}$$

$$\vec{\nabla}\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2kr) = 3k$$



$$\int_V \vec{\nabla}\cdot\vec{F}\;dv = 3k\int_{r_1}^{r_2} r^2dr\int_0^{\pi}\sin\theta\;d\theta\int_0^{2\pi}d\varphi = 4\pi k(r_2^3 - r_1^3)$$

$$\oint_S \vec{F}\cdot\overrightarrow{dS} = \iint_{outer}\vec{F}\cdot\overrightarrow{dS} + \iint_{inner}\vec{F}\cdot\overrightarrow{dS}$$

$$\oint_S \vec{F}\cdot\overrightarrow{dS} = kr_2^3\int_0^{2\pi}d\varphi\int_0^{\pi}\sin\theta\,d\theta - kr_1^3\int_0^{2\pi}d\varphi\int_0^{\pi}\sin\theta\,d\theta = 4\pi k(r_2^3 - r_1^3)$$

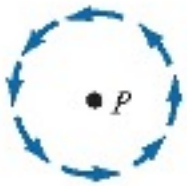
Vector Differentiation: Curl

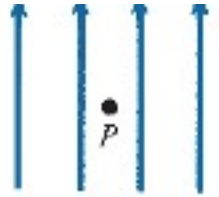
The curl operation measures how strong a vector field "circulates" around a point. The result is a vector that is normal to the circulation path.

Rectangular coordinate

$A(x, y, z)$ is a scalar function

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

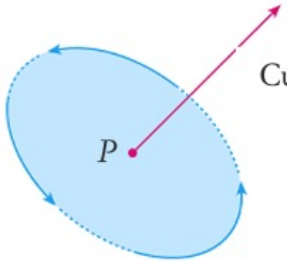




Cylindrical coordinate

$A(\rho, \varphi, z)$ is a scalar function

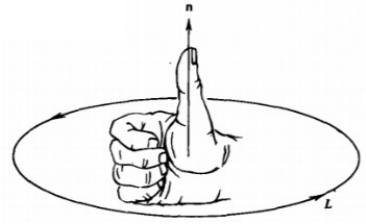
$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$



Spherical coordinate

$A(r, \theta, \varphi)$ is a scalar function

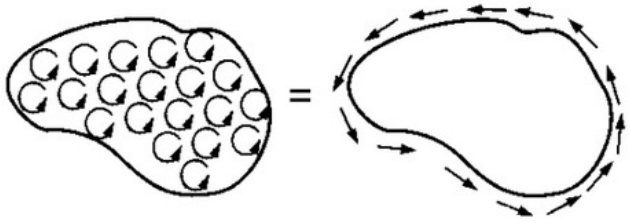
$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \mathbf{a}_r & r\mathbf{a}_\theta & (r\sin\theta)\,\mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & (r\sin\theta)\,A_\phi \end{vmatrix}$$



Stokes Theorem

$$\int_S (\vec{\nabla}\times\vec{A}) \cdot \overrightarrow{dS} = \oint_L \vec{A} \cdot \overrightarrow{dl}$$

Sum of all curl magnitudes of points over a surface is equal to the line integral around the surface boundary.



Examples

(1) Does the vector field $\vec{F} = xy\,\vec{a}_x - 2x\,\vec{a}_y$ exhibit any circulation behavior at point P(-2, 1, 3)?

$$\vec{\nabla}\times\vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -2x & 0 \end{vmatrix} = -(2+x)\,\vec{a}_z \qquad \left(\vec{\nabla}\times\vec{F}\right)\Big|_P = 0$$

(No circulation behavior)

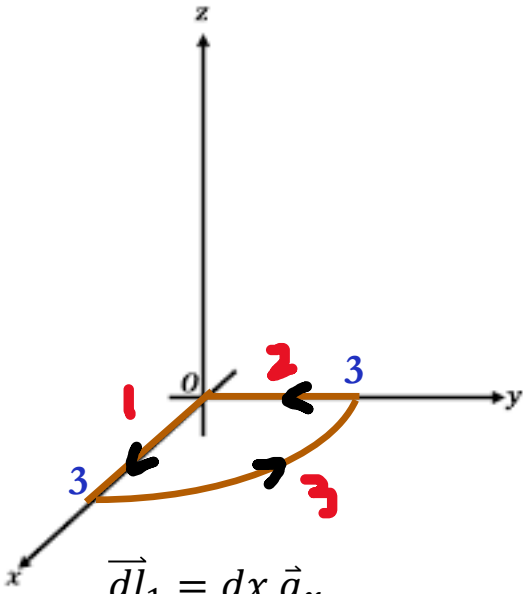
(2) Verify Stokes theorem over the given contour.

$$\int_S \left(\vec{\nabla}\times\vec{F}\right)\cdot\overrightarrow{dS} = \int_0^{\frac{\pi}{2}}\int_0^3 -(2+\rho\cos\varphi)\,\rho\,d\rho\,d\varphi = -9\left(1+\frac{\pi}{2}\right)$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_{L1}\vec{F}\cdot\overrightarrow{dl}_1 + \int_{L2}\vec{F}\cdot\overrightarrow{dl}_2 + \int_{L3}\vec{F}\cdot\overrightarrow{dl}_3$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_{L1} xy\,dx - \int_{L2} 2x\,dy + \int_{L3}(xy\,dx - 2x\,dy)$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_0^3 x\sqrt{9-x^2}\,dx - \int_3^0 2\sqrt{9-y^2}\,dy = -9\left(1+\frac{\pi}{2}\right)$$



$$\overrightarrow{dl}_1 = dx\,\vec{a}_x$$

$$\overrightarrow{dl}_2 = dy\,\vec{a}_y$$

$$\overrightarrow{dl}_3 = dx\,\vec{a}_x + dy\,\vec{a}_y$$

Vector Differentiation: Laplacian

The Laplacian is a combination of operators.

It can operate on scalar distributions. $\nabla^2 A = \vec{\nabla} \cdot \vec{\nabla} A$

In rectangular coordinated: $\nabla^2 A(x,y,z) = \frac{\partial^2}{\partial x^2} A(x,y,z) + \frac{\partial^2}{\partial y^2} A(x,y,z) + \frac{\partial^2}{\partial z^2} A(x,y,z)$

For the case $\nabla^2 A = 0$, the solution for A is sines and cosines (harmonic solution)

It can operate on vector distributions. $\vec{\nabla}^2 \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla} \times \vec{\nabla} \times \vec{A}$

In rectangular coordinated: $\vec{\nabla}^2 \vec{A} = \nabla^2 A_x \, \vec{a}_x + \nabla^2 A_y \, \vec{a}_y + \nabla^2 A_z \, \vec{a}_z$

Example: Given the potential distribution $V(x,y,z) = e^{-z} \sin 2x \cosh y$

$$\nabla^2 V = e^{-z} \cosh y \frac{\partial^2}{\partial x^2} \sin 2x + e^{-z} \sin 2x \frac{\partial^2}{\partial y^2} \cosh y + \sin 2x \cosh y \frac{\partial^2}{\partial z^2} e^{-z}$$

$$\nabla^2 V = -2e^{-z} \sin 2x \cosh y$$

Remarks

(1) Differentiation operators are "point" operators.

(2) A vector field is uniquely specified within a given space by its divergence and curl.

(3) A vector field with $\vec{\nabla} \cdot \vec{A} = 0$, is solenoidal in nature (field lines go in closed paths).

(4) A solenoidal vector field can always be expressed as the curl of another vector function (Example: $\vec{\nabla} \cdot \vec{B} = 0 \rightarrow \vec{F} = \vec{\nabla} \times \vec{B}$).

(5) A vector field with $\vec{\nabla} \times \vec{A} = 0$, is irrotational in nature (also called conservative).

(6) A conservative vector field can always be expressed as the gradient of a scalar function (Example: $\vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E} = -\vec{\nabla} V$).

(7) A scalar function with $\nabla^2 A = 0$, is sinusoidal in nature (sine and cosine variation).

Q & A

Send me an email if you have any questions.